Neural Networks

Technical reference

The general model of a neural network, learning tasks and their error, and data representation, with proofs. Each chapter says which site note explains it; the last chapter collects the formulas.

Companion to the notes at https://lorenzoliuzzo.github.io/notes/

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1 The general model

In the notes: Introduction to Neural Networks

1.1 Networks, units and connections

DEFINITION Neural network A neural network is a directed graph $G = (U, C)$ with a finite set U of *units* and a set $C \subseteq U \times U$ of *connections*, together with a weight $w_{uv} \in \mathbb{R}$ for every connection $(v, u) \in C$. The set of predecessors of a unit u is $\mathrm{pred}(u) = \{v : (v, u) \in C\}$. The units are partitioned into the input units U_{in} and output units U_{out}, both non-empty and not necessarily disjoint, and the hidden units $U_{\mathrm{hid}} = U \setminus (U_{\mathrm{in}} \cup U_{\mathrm{out}})$.

Every unit u has three real state variables, the *network input* net_u, the *activation* act_u and the *output* out_u, and every input unit has a fourth, the *external input* ext_u. Every unit has three functions, with θ_u the vector of its own parameters (for example a threshold):

$$\mathrm{net}_u = f_{\mathrm{net}}^{(u)}(w_u, \mathrm{in}_u), \quad \mathrm{act}_u = f_{\mathrm{act}}^{(u)}(\mathrm{net}_u, \theta_u), \quad \mathrm{out}_u = f_{\mathrm{out}}^{(u)}(\mathrm{act}_u), \tag{1}$$

where $w_u = (w_{uv})_{v \in \mathrm{pred}(u)}$ and $\mathrm{in}_u = (\mathrm{out}_v)_{v \in \mathrm{pred}(u)}$. For an input unit, $\mathrm{net}_u = \mathrm{ext}_u$.

DEFINITION Network input function The function $f_{\mathrm{net}}^{(u)}$ of a unit. The standard choice is the weighted sum $\mathrm{net}_u = \sum_{v \in \mathrm{pred}(u)} w_{uv}\, \mathrm{out}_v$; radial basis function units use a distance $d(w_u, \mathrm{in}_u)$ instead. The activation and output functions are, for the standard threshold unit, a step at θ_u and the identity.

Architectures differ by the graph and by which of the three functions they change: a distance replaces the sum in radial basis function networks, a winner-takes-all rule replaces the identity output in learning vector quantization, and in a Hopfield network $U_{\text{in}} = U_{\text{out}} = U$ and $U_{\text{hid}} = \varnothing$.

1.2 Acyclic networks

DEFINITION Feed-forward network A neural network whose graph has no directed cycle. A network with a cycle is called recurrent.

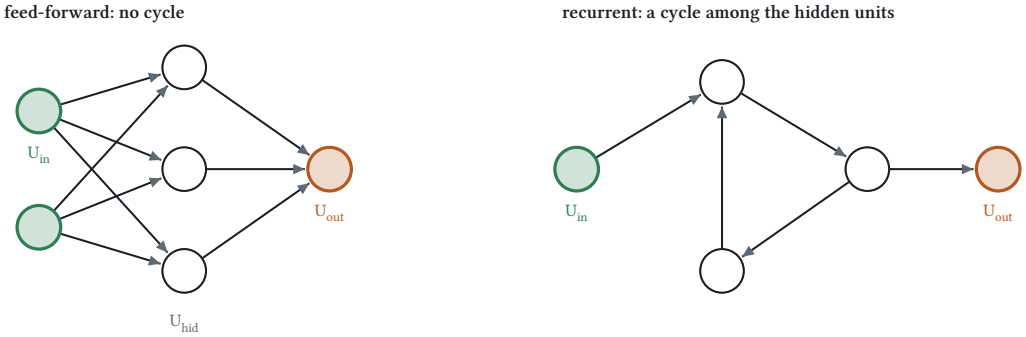
LEMMA A finite directed graph is acyclic if and only if its vertices can be numbered $u_1, \dots, u_n$ so that $(u_j, u_i) \in C$ implies $j < i$ (a *topological order*).

Proof. If such a numbering exists, a cycle $v_1 \to v_2 \to \dots \to v_1$ would give an increasing sequence of numbers returning to its start, which is impossible. Conversely, let G be acyclic. Some vertex has no predecessor: otherwise, starting anywhere and repeatedly moving to a predecessor produces, after $n + 1$ steps, a repeated vertex and therefore a cycle. Give that vertex the number 1, remove it (the rest is still acyclic) and number the remaining vertices by induction. □

THEOREM Evaluation of a feed-forward network Let the external inputs be fixed and the units be taken in a topological order. Computing for each unit in turn net_u, act_u and out_u determines every output after $|U|$ unit evaluations, and the result does not depend on which topological order is used.

Proof. By induction on the position i in the order: unit u_i depends only on the weights, on θ_{u_i} and on the outputs of its predecessors, all of which have position below i and are already determined (an input unit depends only on its external input). So the value of out_{u_i} is a function of the data alone, defined by the same recursion for every topological order. □

Remark. A recurrent network has no such order. After the input phase, with the external input switched off, units are recomputed whenever one of their inputs has changed, until the outputs are stable or a bound on the number of recomputations is reached. Both the outcome and whether a stable state exists can depend on the update order. For example, two units u, v with $\text{out}_u := \text{out}_v$ and $\text{out}_v := 1 - \text{out}_u$ and outputs in $\{0, 1\}$ pass through $(0,0) \to (0,1) \to (1,1) \to (1,0) \to (0,0)$ and never stabilize.



Left: one pass in topological order computes everything. Right: the outputs must be recomputed until they stop changing.

Figure 1: An acyclic graph (left) and a graph with a cycle (right). Only the first can be evaluated by a single pass in topological order.

2 Learning tasks

In the notes: Introduction to Neural Networks

2.1 Fixed and free tasks

DEFINITION Fixed learning task For a network with input units $u_1, \dots, u_n$ and output units $v_1, \dots, v_m$, a fixed learning task is a finite set $L = \{(\iota^{(l)}, o^{(l)})\}_{l=1}^{N}$ of training patterns, each an input vector $\iota^{(l)} \in \mathbb{R}^n$ (the external inputs) and a desired output vector $o^{(l)} \in \mathbb{R}^m$. The task is solved if for every l the network, given $\iota^{(l)}$, computes $\mathrm{out}_{v_k} = o_k^{(l)}$ for all k. It is the supervised learning task.

DEFINITION Free learning task A free learning task is a finite set $L = \{\iota^{(l)}\}$ of input vectors with no desired outputs. The outputs are chosen by the learning method, under the principle that similar inputs should produce similar outputs.

DEFINITION Sum of squared errors The error of a network on a fixed learning task is

$$e = \sum_{l=1}^{N} \sum_{k=1}^{m} \left(o_k^{(l)} - \mathrm{out}_{v_k}^{(l)} \right)^2, \tag{2}$$

where $\mathrm{out}_{v_k}^{(l)}$ is the output of v_k for input $\iota^{(l)}$. It is non-negative, and it is zero exactly when the task is solved.

2.2 The error as maximum likelihood

THEOREM Squared error and Gaussian noise Suppose the desired outputs satisfy $o^{(l)} = y(\iota^{(l)}; w) + \varepsilon^{(l)}$, where $y(\cdot; w)$ is the function computed by the network with parameters w and the $\varepsilon^{(l)}$ are independent with distribution $\mathcal{N}(0, \sigma^2 I_m)$. Then the log-likelihood of the data is

$$\ell(w, \sigma^2) = -\frac{e(w)}{2\sigma^2} - \frac{Nm}{2} \log(2\pi\sigma^2). \tag{3}$$

For every fixed σ^2 it is maximized by the w that minimizes e, and the maximum likelihood estimate of the variance is $\hat{\sigma}^2 = e(\hat{w})/(Nm)$.

Proof. By independence the density of the data is $\prod_{l,k} (2\pi\sigma^2)^{-1/2} \exp\left(-\left(o_k^{(l)} - y_{k(\iota^{(l)};w)} \right)^2 / (2\sigma^2) \right)$; taking logarithms gives the stated ℓ. The first term is a decreasing function of $e(w)$ for fixed σ^2, which gives the claim about w. Setting $\partial \ell / \partial \sigma^2 = e/(2\sigma^4) - Nm/(2\sigma^2) = 0$ gives $\hat{\sigma}^2 = e/(Nm)$. □

Remark. For class labels the Gaussian model is replaced by a Bernoulli or categorical one, whose negative log-likelihood is the cross-entropy; see Bishop (2006), section 5.2.

3 Data representation

In the notes: Introduction to Neural Networks

DEFINITION Standardization For an input with values $x_1, \dots, x_n$, mean $\bar{x} = 1/n \sum_i x_i$ and corrected standard deviation $s = \sqrt{1/(n-1) \sum_i (x_i - \bar{x})^2} > 0$, the standardized values are $x_i' = (x_i - \bar{x})/s$.

THEOREM The standardized values have mean 0 and corrected standard deviation 1.

Proof. $1/n \sum_i x_i' = (\bar{x} - \bar{x})/s = 0$ and $1/(n-1) \sum_i x_i'^2 = 1/s^2 \cdot 1/(n-1) \sum_i (x_i - \bar{x})^2 = 1$. □

The mean and s are computed on the training data and the same numbers are applied to validation and test data.

DEFINITION **One-hot encoding** A variable with symbols $a_1, \ldots, a_N$ is coded by N input units; symbol a_i becomes the vector $e_i \in \{0,1\}^N$ with a 1 in position i and 0 elsewhere (called 1-in-N encoding in the lectures).

THEOREM For $i \neq j$, $\|e_i - e_j\|_2 = \sqrt{2}$. In contrast, coding the symbols by integers z_i gives distances $|z_i - z_j|$ that differ between pairs, and $|z_1 - z_3| = |z_1 - z_2| + |z_2 - z_3|$ whenever z_2 lies between z_1 and z_3.

Proof. $e_i - e_j$ has a 1 in position i, a -1 in position j and zeros elsewhere, so its squared norm is 2. The statement on integer codes is the additivity of distance on a line. □

Remark. Standardization and decorrelation of the inputs are recommended for gradient-based training by LeCun et al. (1998); the argument is about the conditioning of the error surface and is not proved here.

4 Summary of formulas

QUANTITY	RESULT	WHERE EXPLAINED
Network input, weighted sum	$\mathrm{net}_u = \sum_{v \in \mathrm{pred}(u)} w_{uv}\, \mathrm{out}_v$	Introduction
Unit state	$\mathrm{net}_u \to \mathrm{act}_u = f_{\mathrm{act}(\mathrm{net}_u, \theta_u)} \to \mathrm{out}_u = f_{\mathrm{out}(\mathrm{act}_u)}$	Introduction
Feed-forward evaluation	one pass in topological order, $\lvert U \rvert$ unit evaluations	Introduction
Sum of squared errors	$e = \sum_l \sum_k \left(o_k^{(l)} - \mathrm{out}_{v_k}^{(l)} \right)^2$	Introduction
Gaussian log-likelihood	$\ell = -e/(2\sigma^2) - (Nm/2)\log(2\pi\sigma^2)$	Introduction
Standardization	$x' = (x - \bar{x})/s$, mean 0, deviation 1	Introduction
One-hot distance	$\|e_i - e_j\| = \sqrt{2}$	Introduction

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